Network 1

Sections 1:3

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Section 1: Installing Programs and Introduction

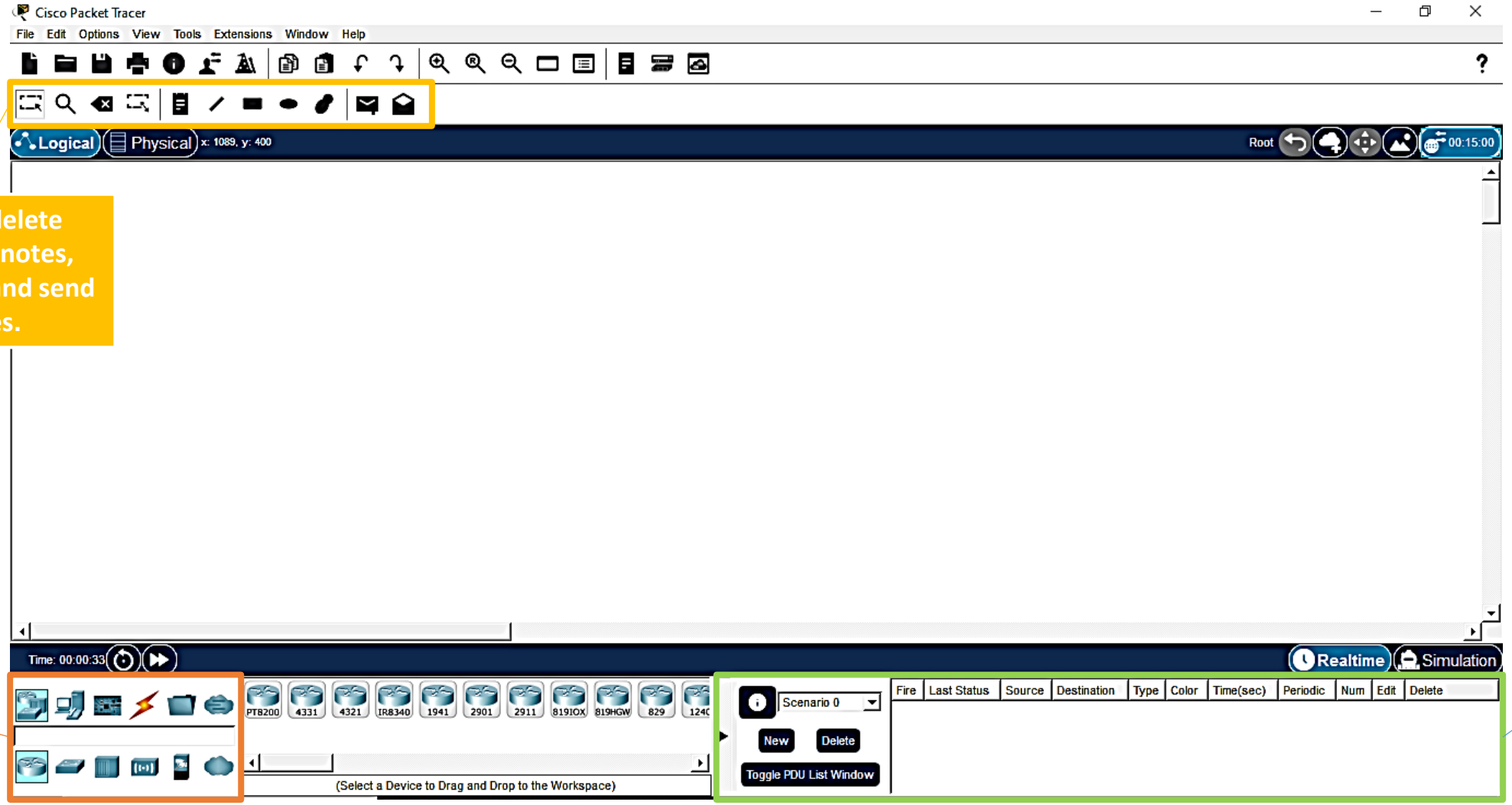
Cisco Packet Tracer

- Instead of using real routers and switches (which are expensive), Packet Tracer lets you simulate a network on your computer.
- You can:
  - Add routers
  - Add switches
  - Connect PCs
  - Configure IP addresses
  - Test routing protocols
  - Simulate DHCP, DNS, HTTP, etc.

Cisco Packet Tracer installation

1) Create a Free Cisco Account https://www.netacad.com/
2) Go to Resources → Download Packet Tracer
   - Choose your operating system:
     - Windows 64-bit or 32-bit
     - Linux
     - macOS
3) Open the downloaded file
4) Next → Accept → Next → Install

Cisco Packet Tracer



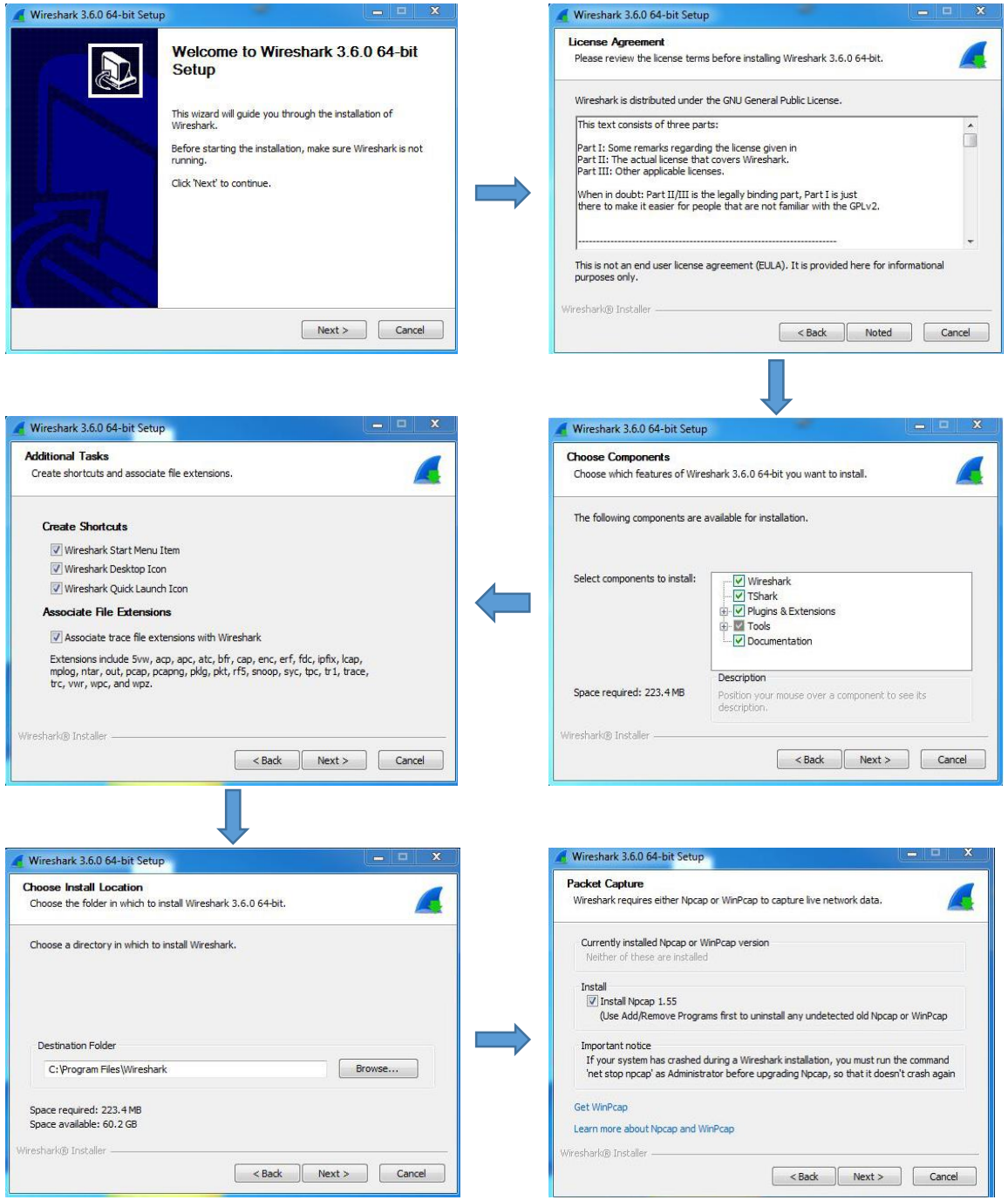
Select and delete devices, add notes, draw shapes, and send messages.

Network devices that can be used

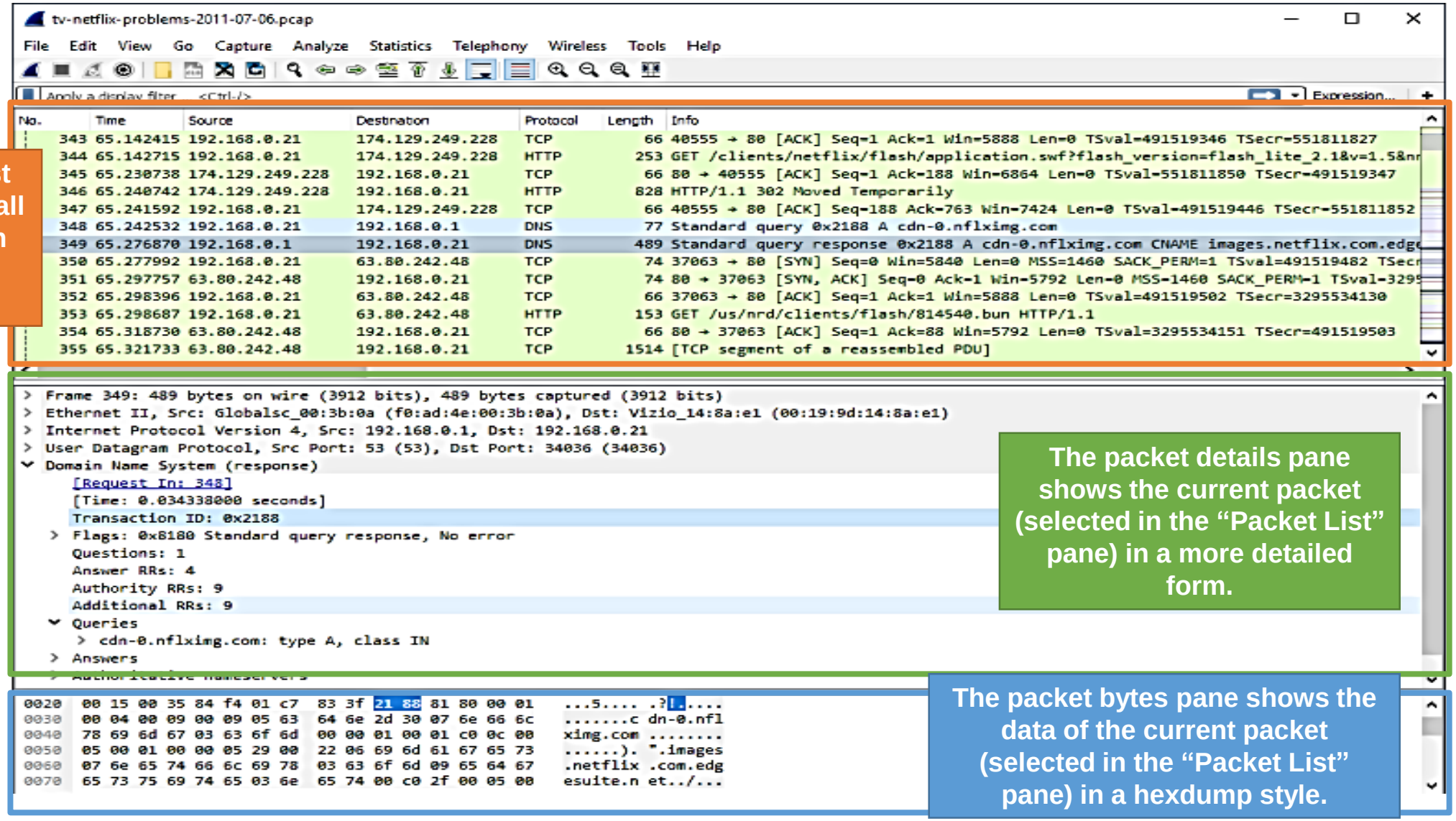
The current user scenario in detail

Wireshark

- Wireshark is a free network protocol analyzer used to capture and analyze network traffic (packets) on your computer
- Download Wireshark
  - Go to the official website: https://www.wireshark.org
  - Click Download.
  - Choose Windows Installer (64-bit).
- Run the Installer:
  - During installation, Wireshark will ask to install Npcap.
  - Next → Next → Next → Install → Finish



Wireshark



The packet list pane displays all the packets in the current capture file.

The packet details pane shows the current packet (selected in the "Packet List" pane) in a more detailed form.

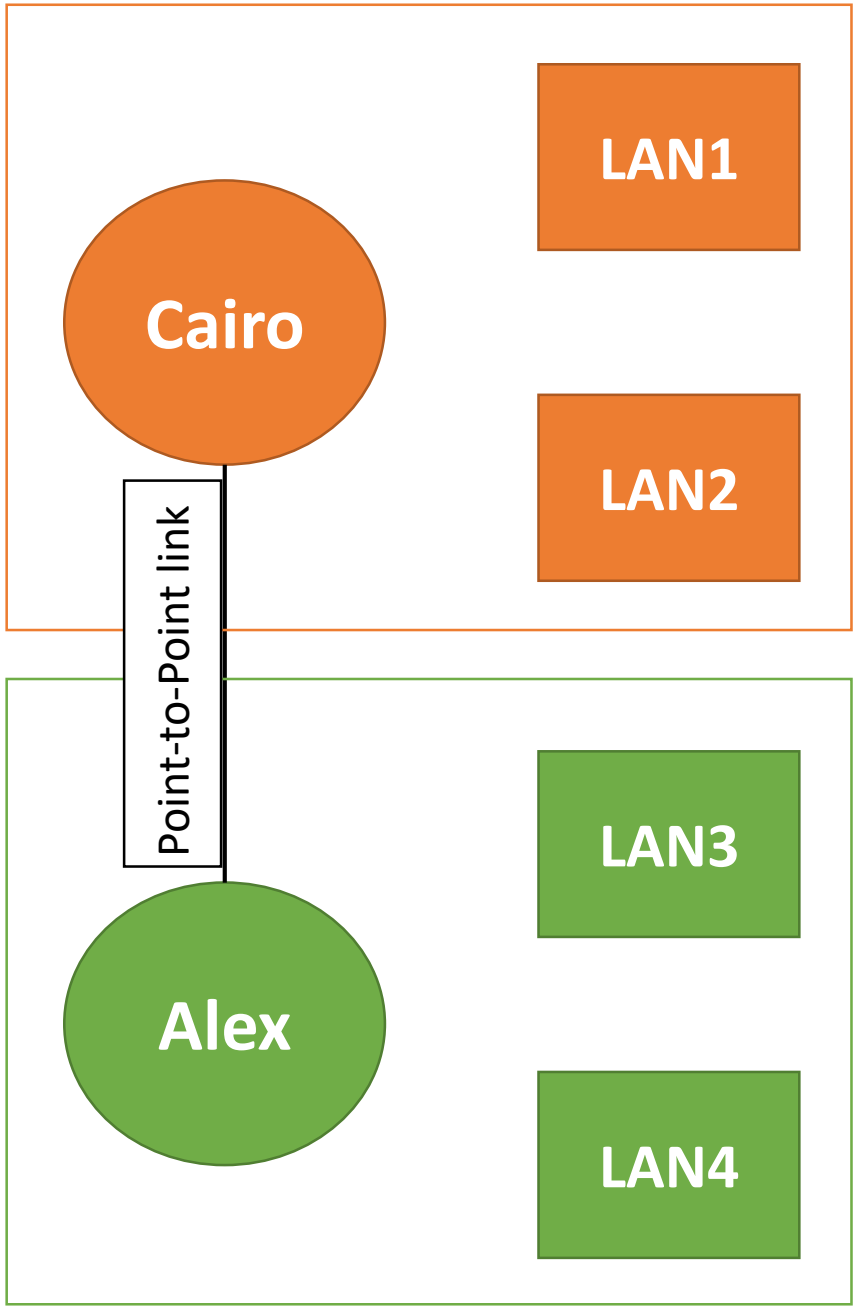
The packet bytes pane shows the data of the current packet (selected in the "Packet List" pane) in a hexdump style.

Section 2: Static VLSM Subnetting

Static VLSM Subnetting

- Variable Length Subnet Masking (VLSM) is an efficient IP addressing technique that allows for creating subnets of different sizes, preventing address waste.
- Unlike FLSM (Fixed Length), VLSM minimizes wasted IP addresses, crucial for IPv4 conservation.
- Example: Company has 2 branches, Cairo has two LANs and Alex has two LANs and P2P between Cairo and Alex.
- You have a single network: 192.168.1.0/24
  - (256 addresses total)
- You need to subnet it Starting from 192.168.1.0 using VLSM for:
  - LAN 1 → 100 hosts (Cairo)
  - LAN 2 → 4 hosts (Cairo)
  - LAN 3 → 30 hosts (Alex)
  - LAN 4 → 42 hosts (Alex)
  - Point-to-Point link between Cairo & Alex routers → 2 hosts

$2^n-2>=$ hosts



Subnet Allocation (Largest to Smallest)

LAN1 (100 hosts) 1
Network: 192.168.1.0/25
Mask: 255.255.255.128
Usable: 192.168.1.1 – 192.168.1.126
Broadcast: 192.168.1.127
Router IP: 192.168.1.1

LAN2 (4 hosts) 4
Network: 192.168.1.224/29
Mask: 255.255.255.248
Usable: 192.168.1.225 – 192.168.1.230
Broadcast: 192.168.1.231
Router IP: 192.168.1.225

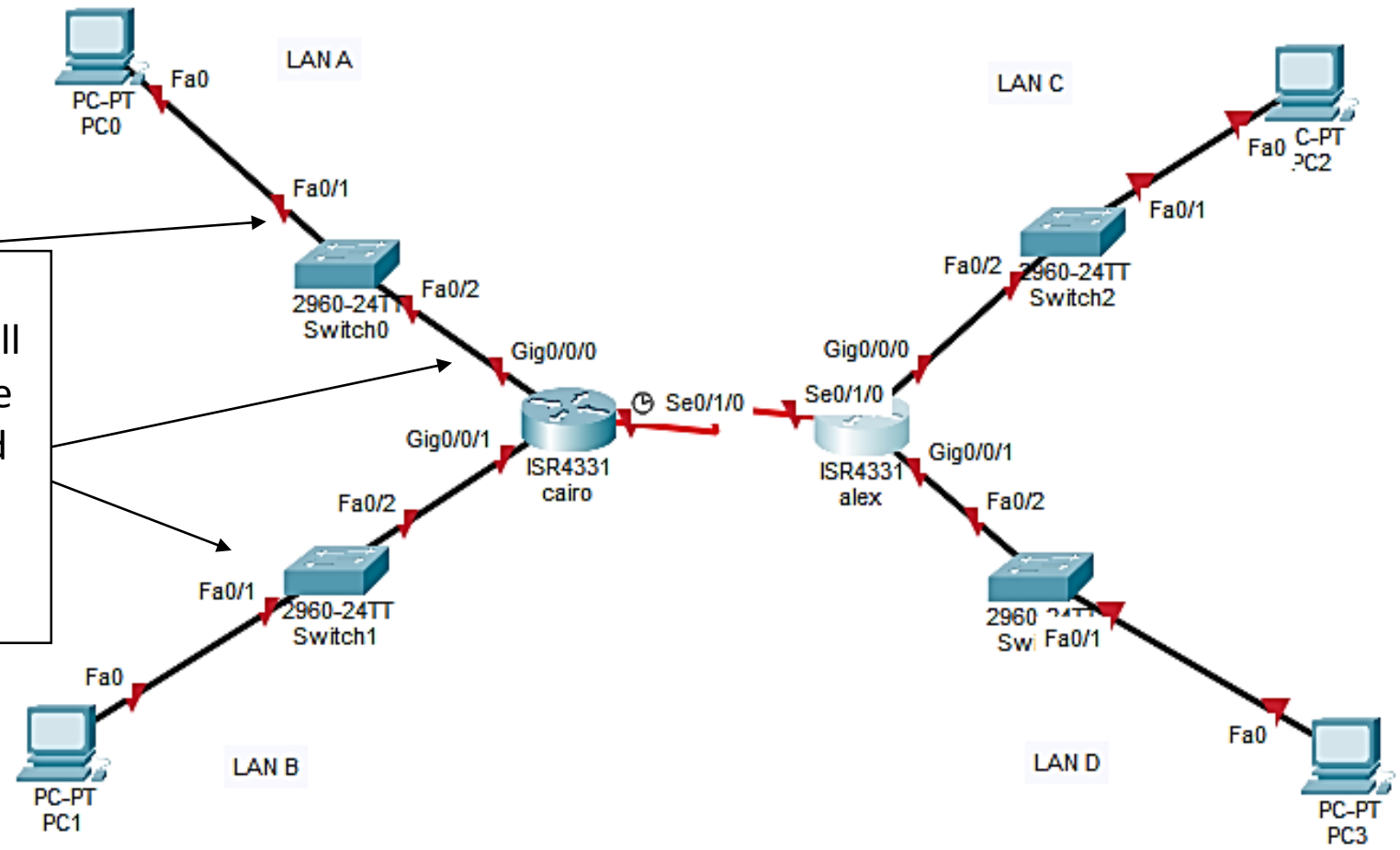
P2P Link 5
Network: 192.168.1.232/30
Mask: 255.255.255.252
Cairo Router: 192.168.1.233
Alex Router: 192.168.1.234
Broadcast: 192.168.1.235

LAN3 (30 hosts) 3
Network: 192.168.1.192/27
Mask: 255.255.255.224
Usable: 192.168.1.193 – 192.168.1.222
Broadcast: 192.168.1.223
Router IP: 192.168.1.193

LAN4 (42 hosts) 2
Network: 192.168.1.128/26
Mask: 255.255.255.192
Usable: 192.168.1.129 – 192.168.1.190
Broadcast: 192.168.1.191
Router IP: 192.168.1.129

Packet Tracer Simulation

Before configuration, all connections are failed (**red**) and turned into (**green**) with configuration

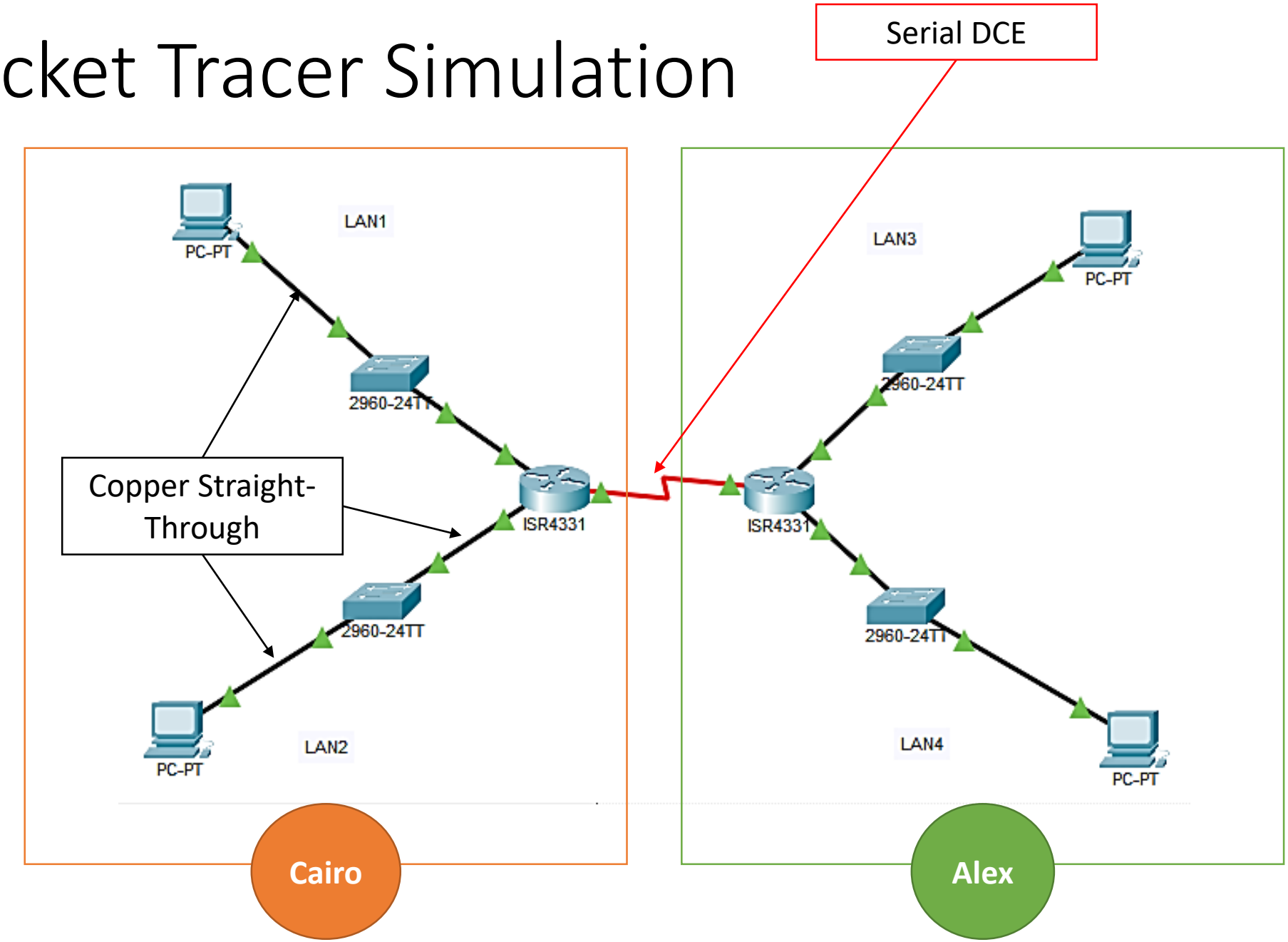


Router

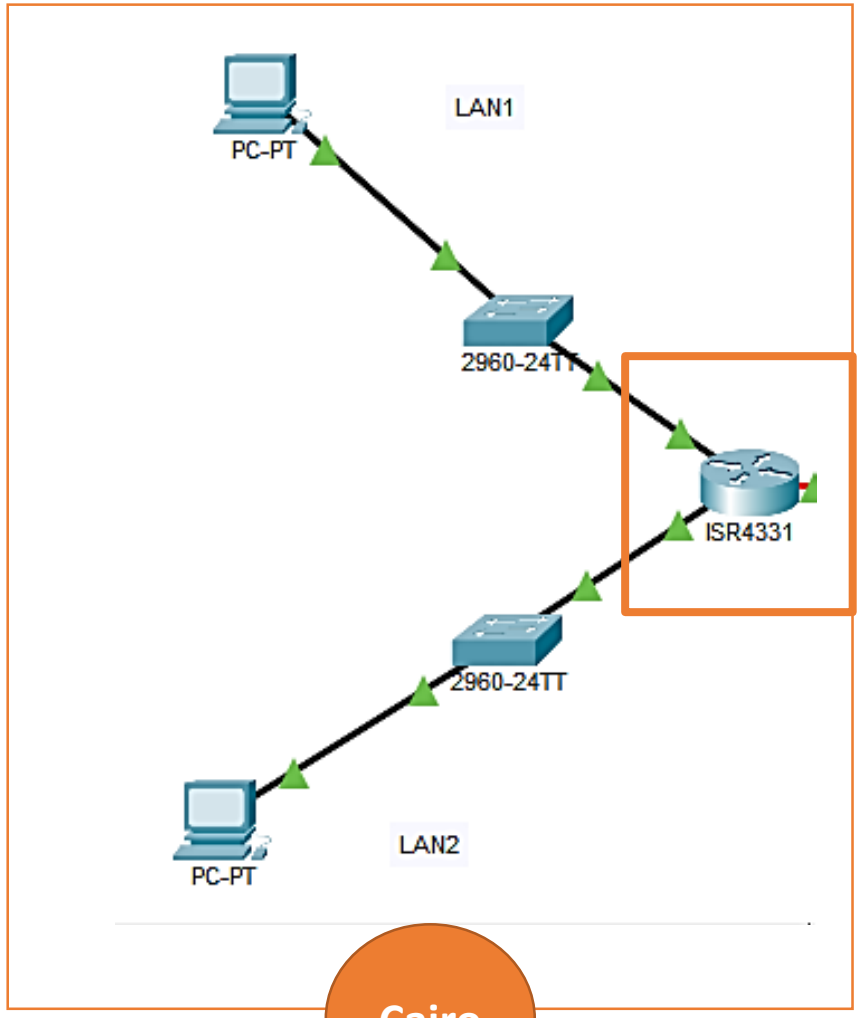
PC

Switch

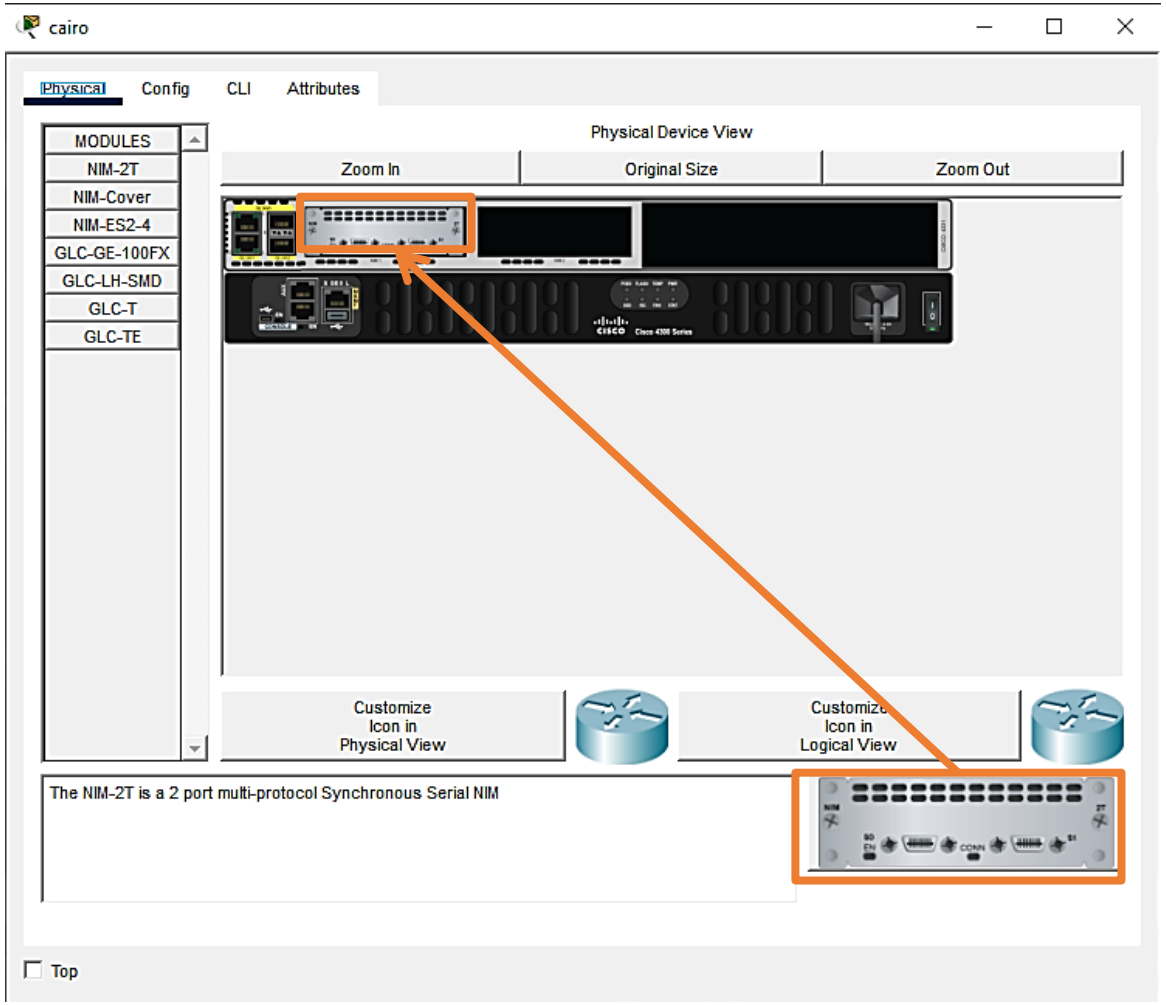
Packet Tracer Simulation



Static Routing (Packet Tracer)



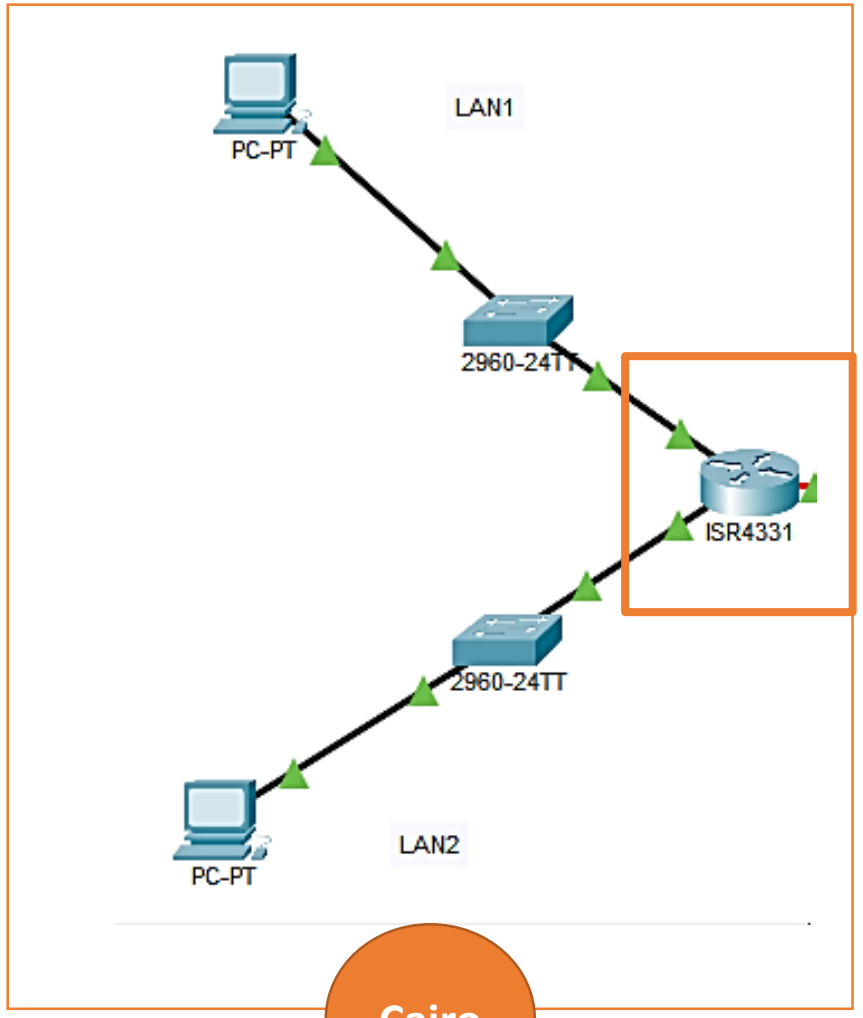
Cairo



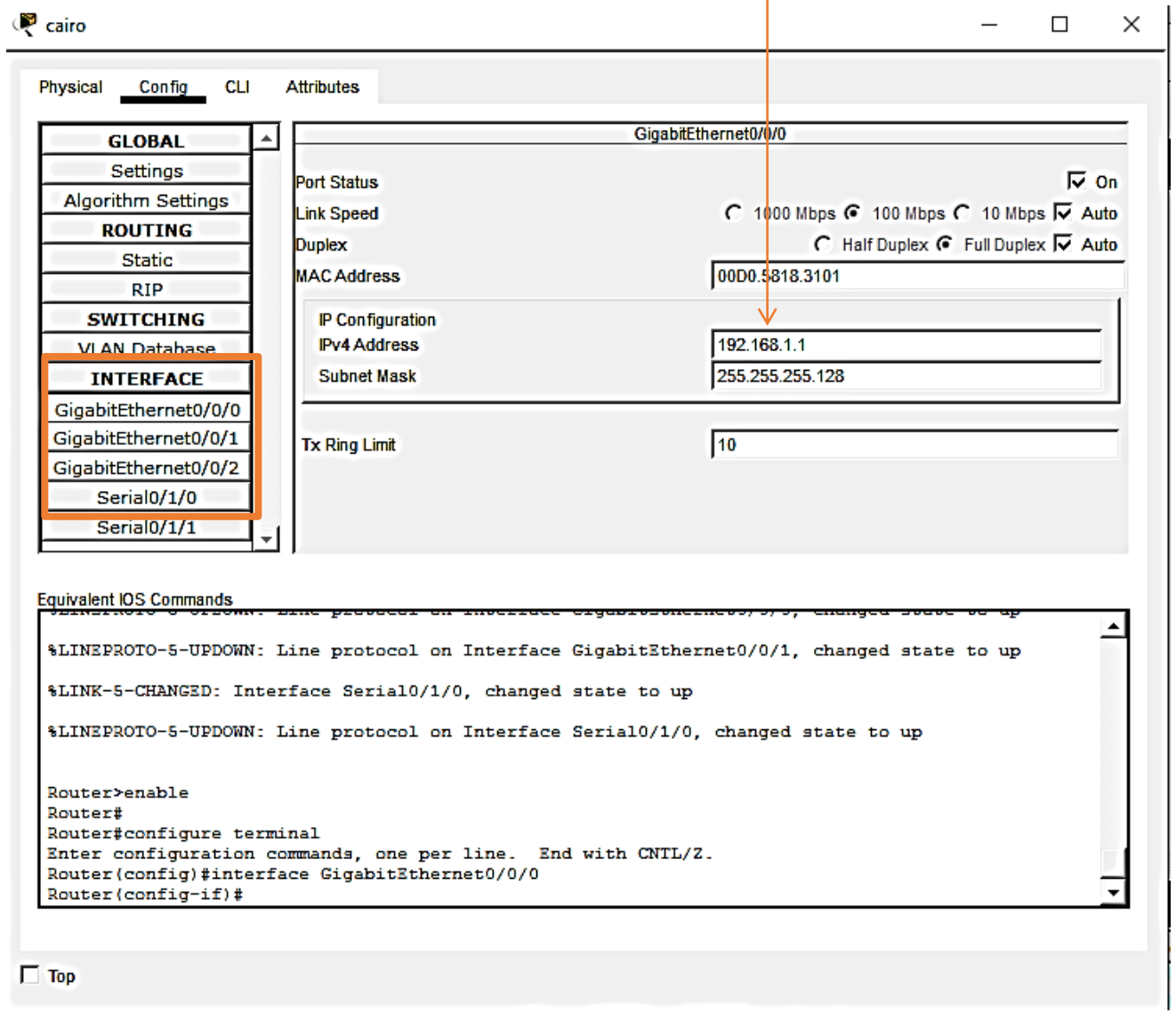
Add serial Serial NIM interface to each routers.
Note: turn off routers before adding

Static Routing (Packet Tracer)

Add IP and Subnet Mask to each interface and turn on port status.

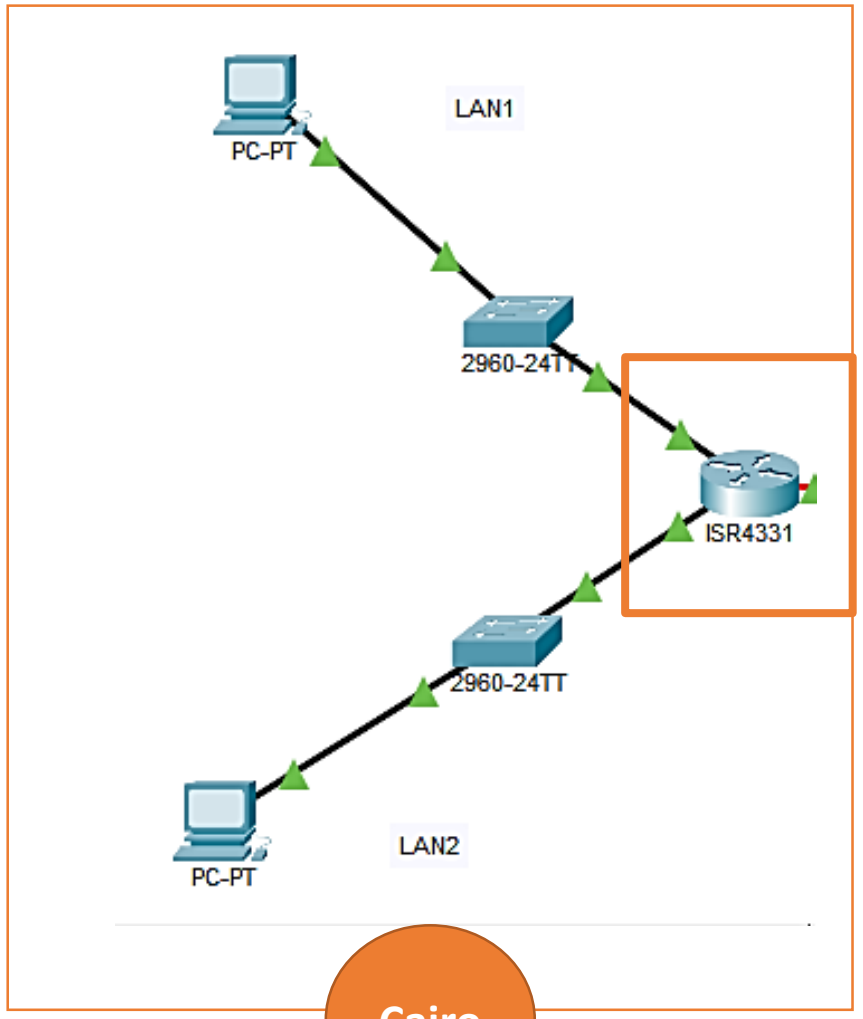


Cairo

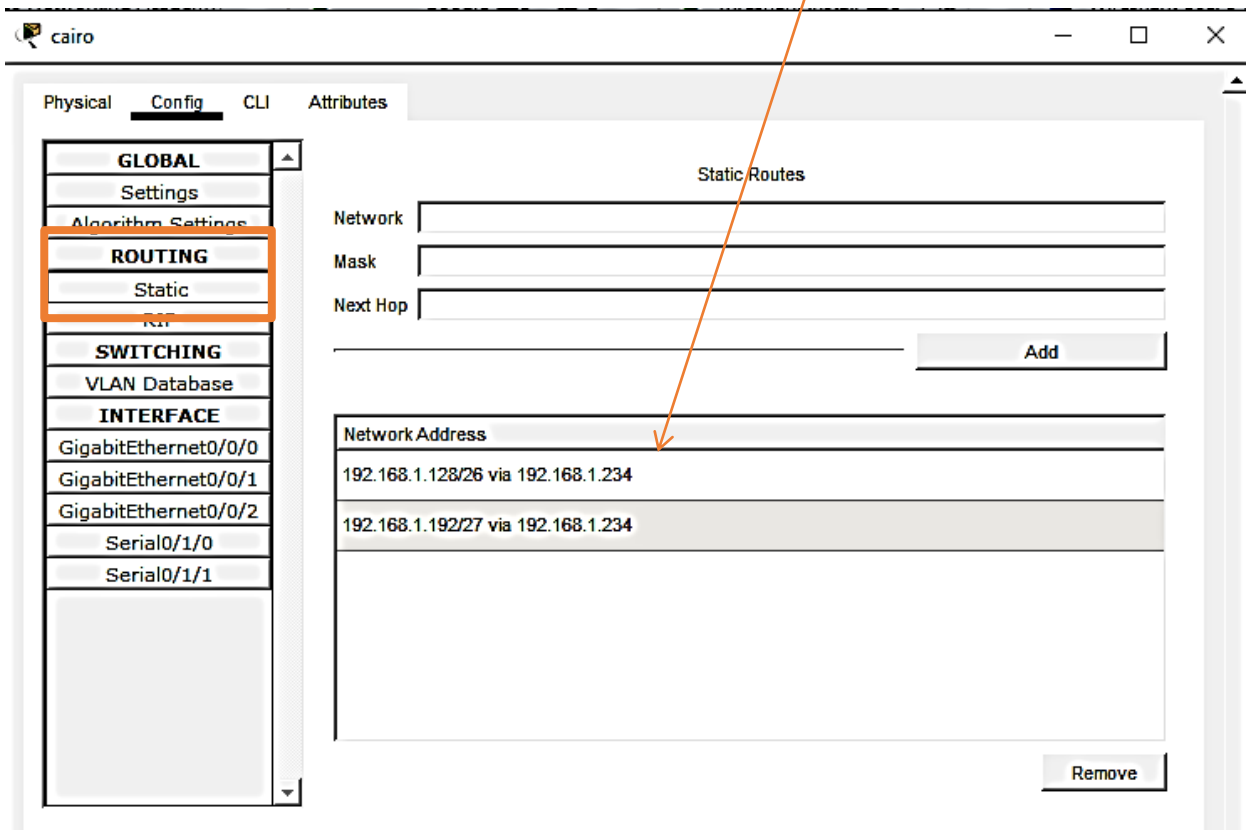


Static Routing (Packet Tracer)

Add Static Routing to each router to see other networks in another router.

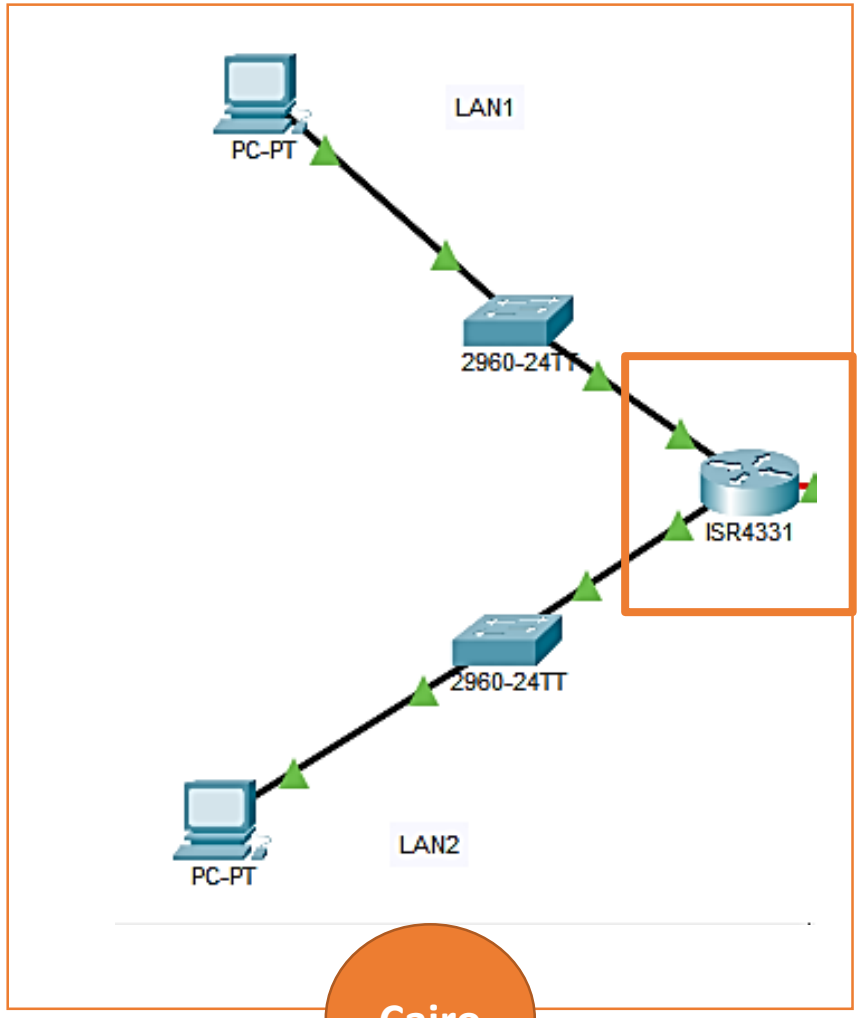


Cairo



Static Routing (Packet Tracer)

You can add IP and Subnet Mask to each interface, turn on port status and add static routing to each router to see other networks in another router using CLI commands.



Cairo

cairo

Physical Config CLI Attributes

IOS Command Line Interface

```
enable
conf t

interface g0/0/0
ip address 192.168.1.1 255.255.255.128
no shutdown

interface g0/0/1
ip address 192.168.1.225 255.255.255.248
no shutdown

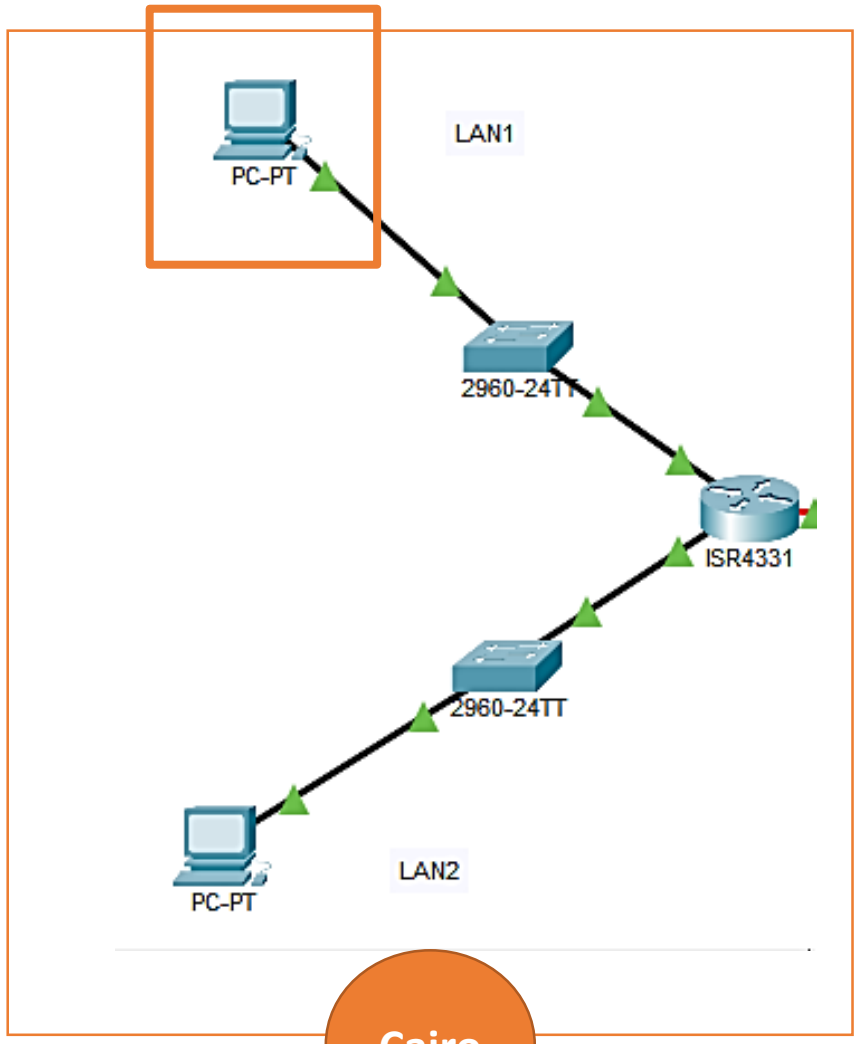
interface s0/1/0
ip address 192.168.1.233 255.255.255.252
no shutdown

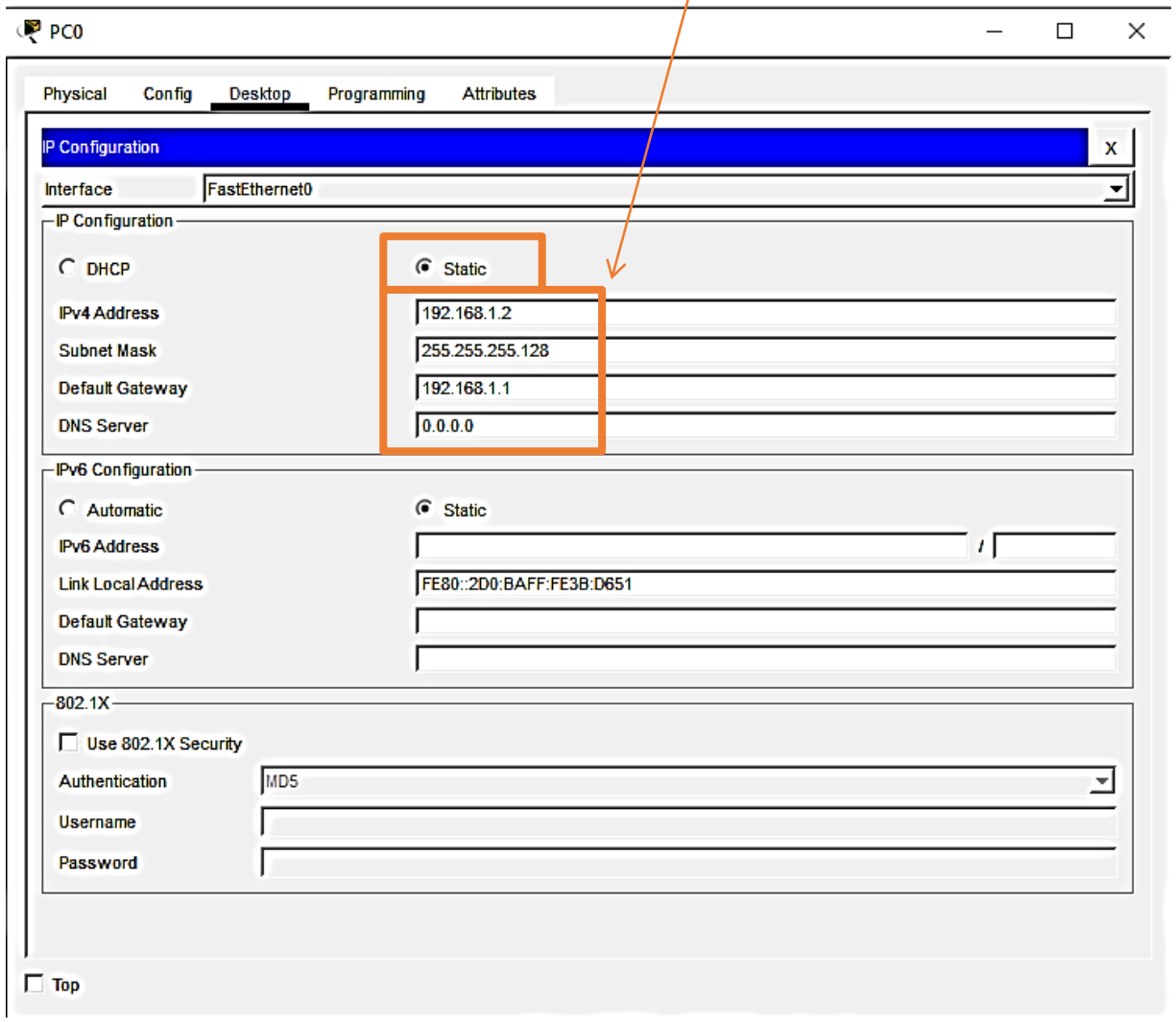
ip route 192.168.1.128 255.255.255.192 192.168.1.234
ip route 192.168.1.192 255.255.255.224 192.168.1.234
```

Top

PC Configurations

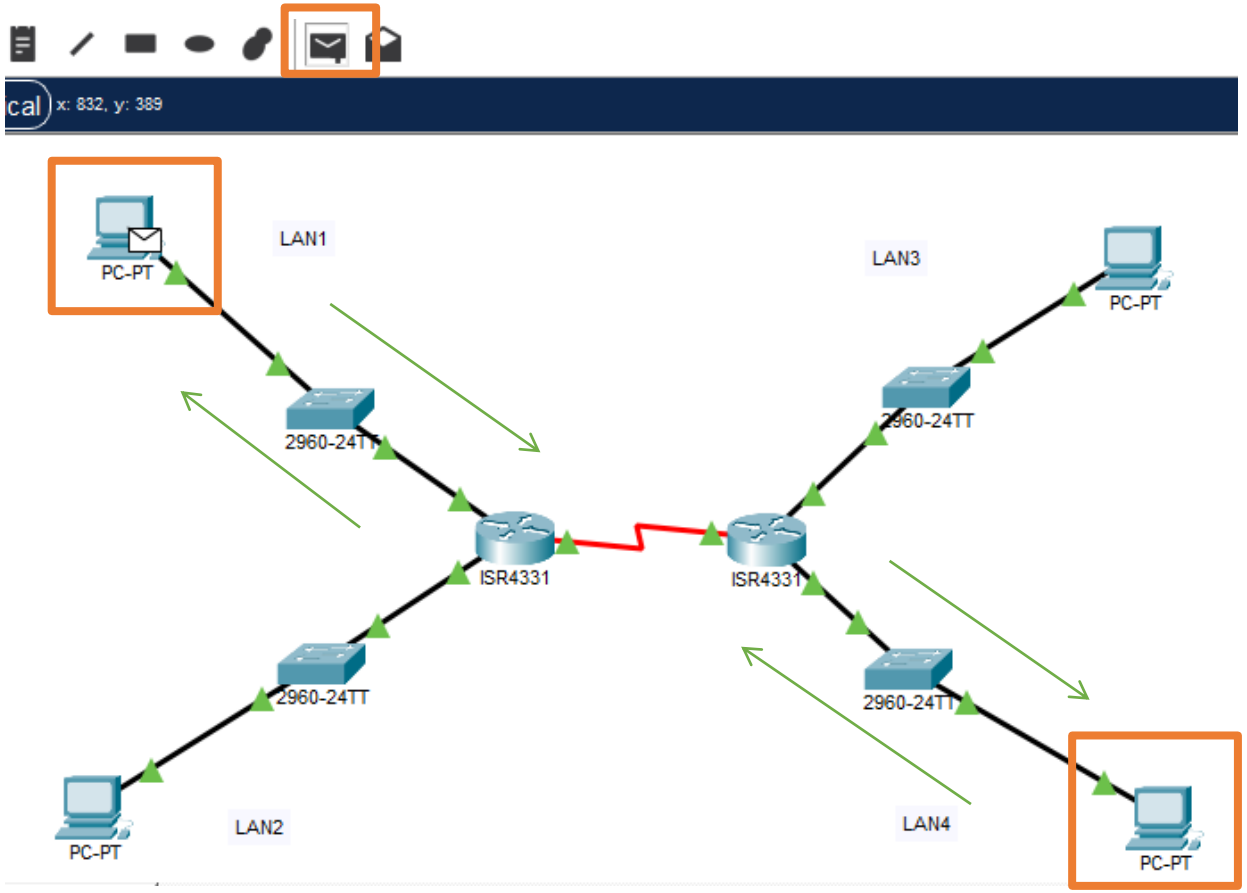
You can add IP and Subnet Mask to each PC, and gateway (router interface IP connected)



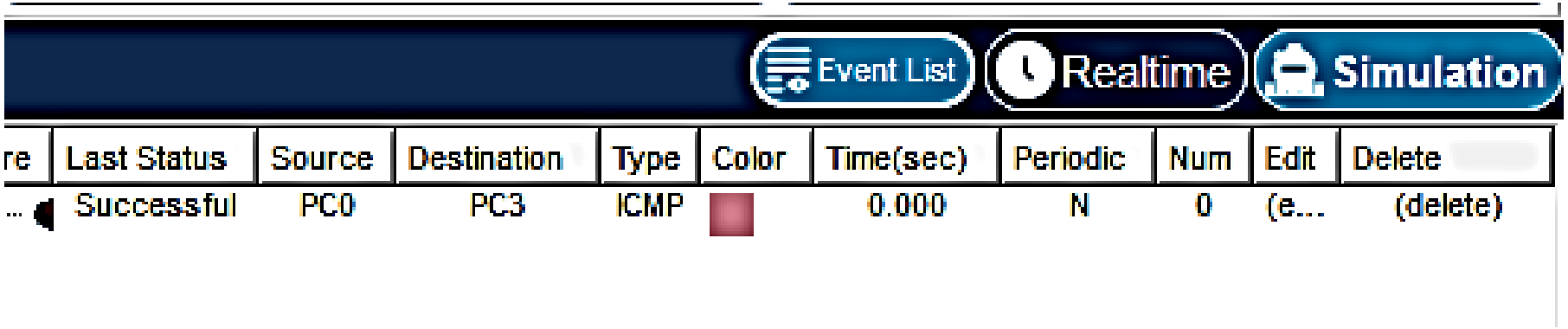


Sending Message

- After inserting all previous steps to all PCs and Routers manually.
- You can send a message from any PC to another by **ping** command in CMD or add simple PDU as figure:



Message Info



re	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
...	Successful	PC0	PC3	ICMP		0.000	N	0	(e...	(delete)

Section 3: DHCP

DHCP

- Dynamic Host Configuration Protocol (DHCP) is a protocol that automatically assigns IP addresses and other network settings to devices in a network.
- Instead of configuring every PC manually (static IP), DHCP gives devices the following automatically when they connect to the network:
  - IP address
  - Subnet mask
  - Default gateway
  - DNS server
- The Dynamic Host Configuration Protocol (DHCP) assigns IP addresses automatically using a 4-step process called DORA.
- This process happens when a device joins the network and needs an IP address.

DHCP

- DORA stands for:
  - D – Discover: PC sends a broadcast message to find a DHCP server.
  - O – Offer: Router suggests an IP address and network configuration
  - R – Request: The client (PC) replies again with Request (accept).
  - A – Acknowledge: It sends ACK containing the full configuration.

In Cisco Packet Tracer when a PC in Cairo LAN1 starts:

1) PC → Discover
2) Cairo Router → Offer (192.168.1.2)
3) PC → Request
4) Router → ACK

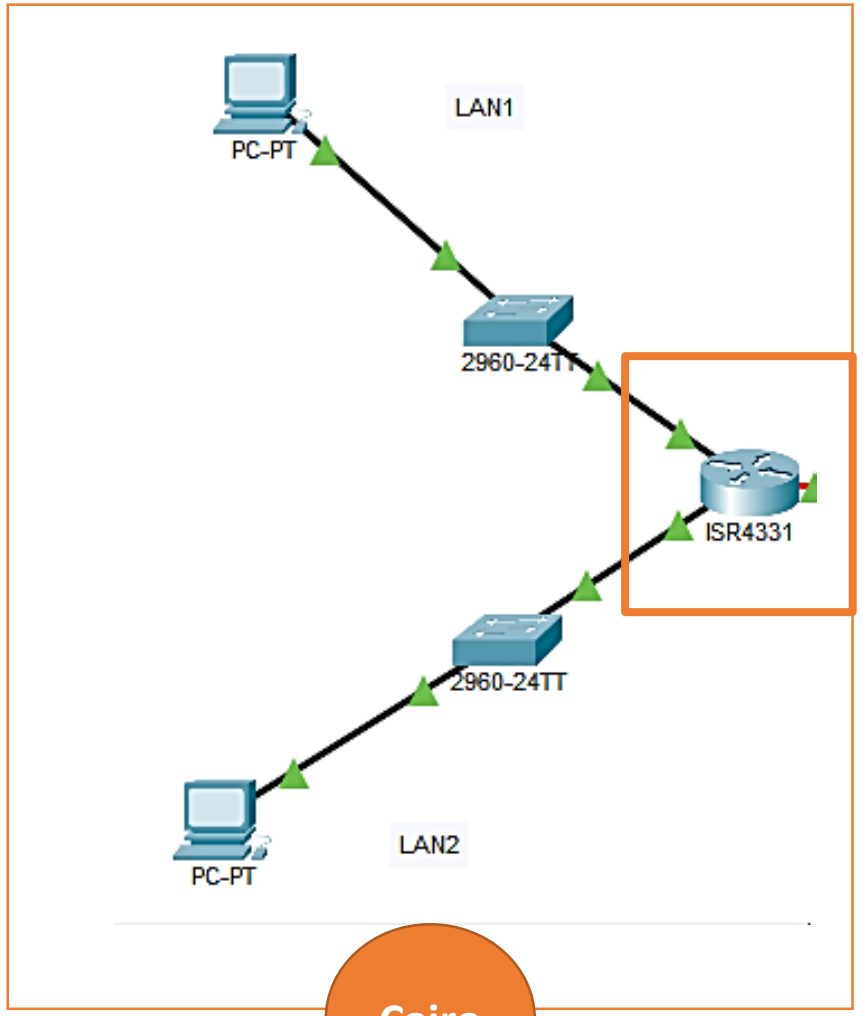
IP: 192.168.1.2
Mask: 255.255.255.128
Gateway: 192.168.1.1

Subnetting using DHCP (Same as Before)

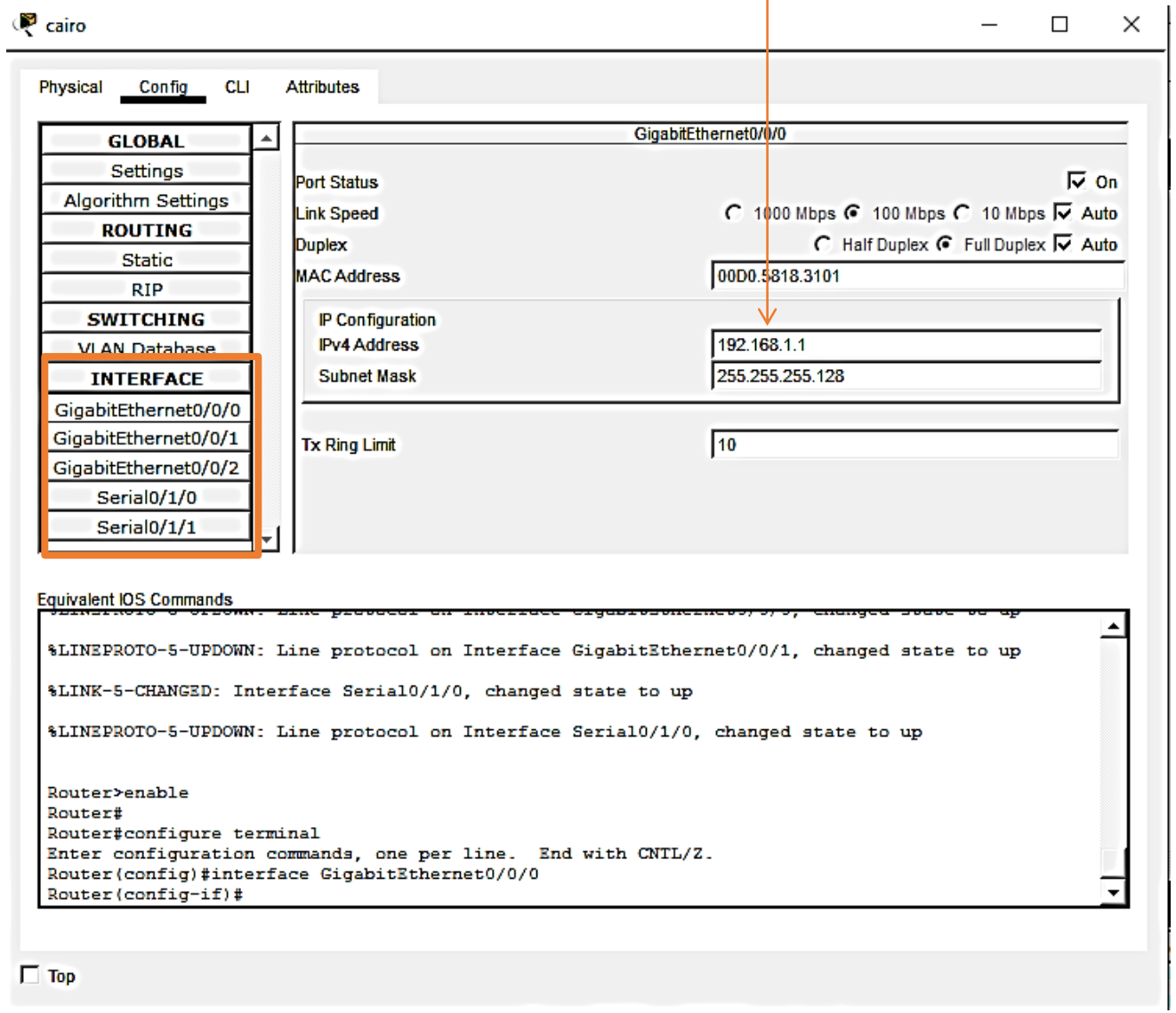
LAN	Network	Mask	Gateway
Cairo LAN1	192.168.1.0/25	255.255.255.128	192.168.1.1
Alex LAN4	192.168.1.128/26	255.255.255.192	192.168.1.129
Alex LAN3	192.168.1.192/27	255.255.255.224	192.168.1.193
Cairo LAN2	192.168.1.224/29	255.255.255.248	192.168.1.225
P2P	192.168.1.232/30	255.255.255.252	routers

Configure Router Interfaces

Add IP and Subnet Mask to each interface and turn on port status.

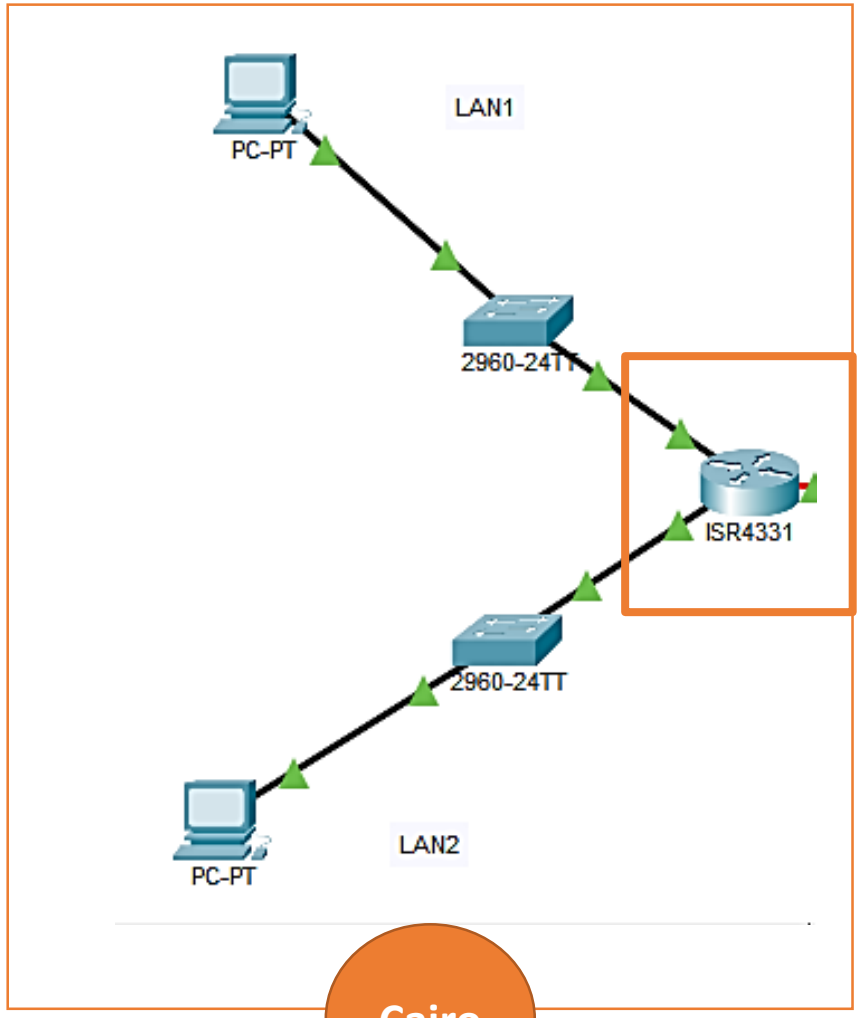


Cairo



You can add IP and Subnet Mask to each interface, turn on port status using CLI commands.

Configure Router Interfaces



Cairo

cairo

Physical Config CLI Attributes

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conf t

interface g0/0/0
ip address 192.168.1.1 255.255.255.128
no shutdown

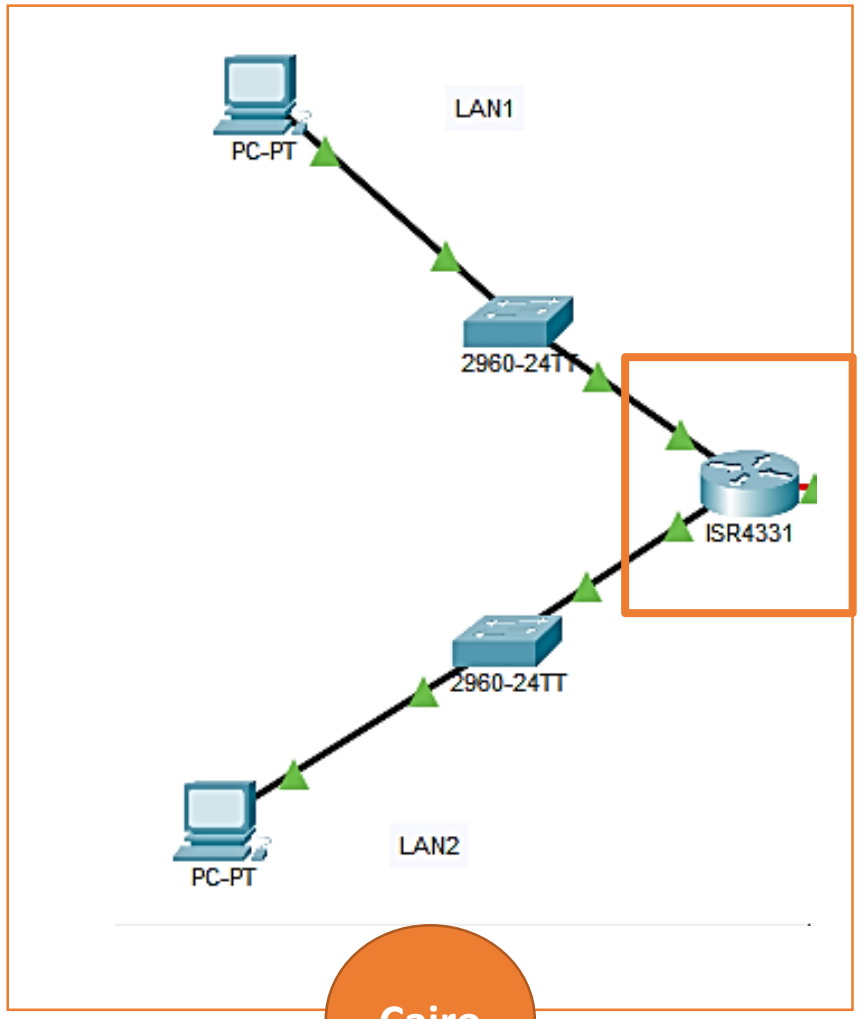
interface g0/0/1
ip address 192.168.1.225 255.255.255.248
no shutdown

interface s0/0/0
ip address 192.168.1.233 255.255.255.252
no shutdown
```

Copy Paste

Top

Configure DHCP



Cairo

cairo

Physical Config CLI Attributes

IOS Command Line Interface

```
ip dhcp excluded-address 192.168.1.1
ip dhcp excluded-address 192.168.1.225

ip dhcp pool LAN1
network 192.168.1.0 255.255.255.128
default-router 192.168.1.1

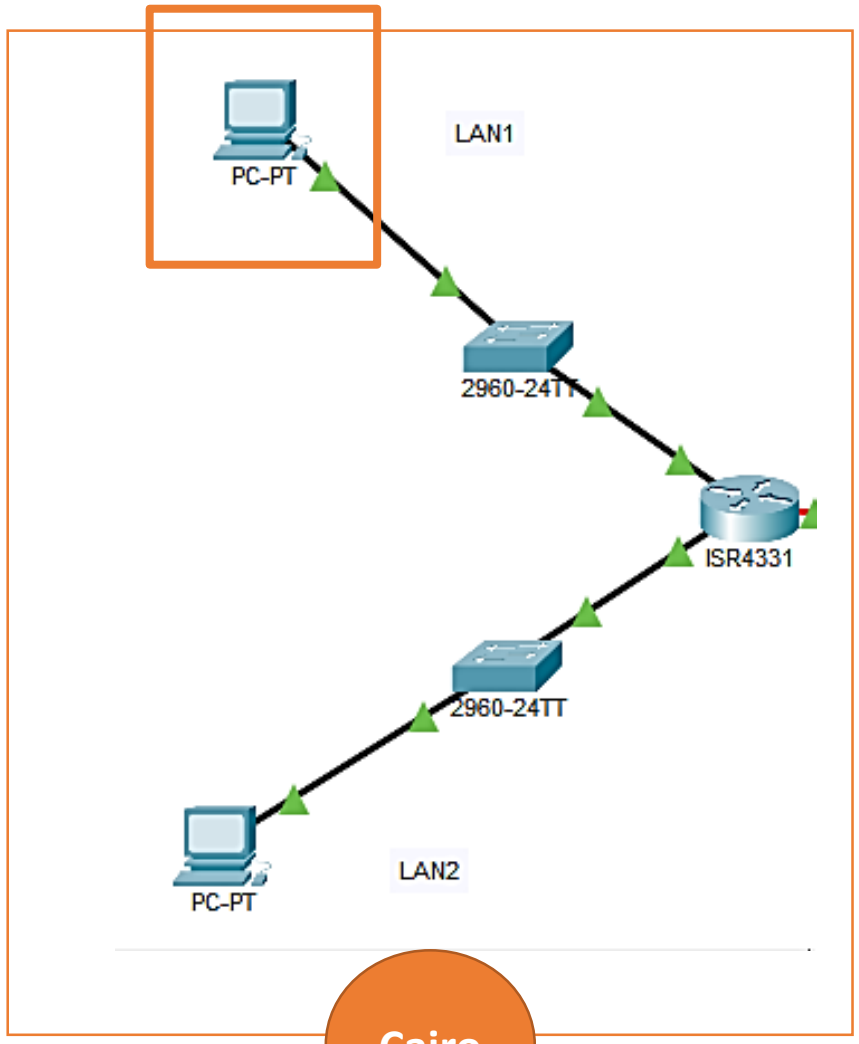
ip dhcp pool LAN2
network 192.168.1.224 255.255.255.248
default-router 192.168.1.225
```

Top

Configure PCs

In Cisco Packet Tracer:

- Click the PC
- Go to Desktop
- Open IP Configuration
- Select DHCP



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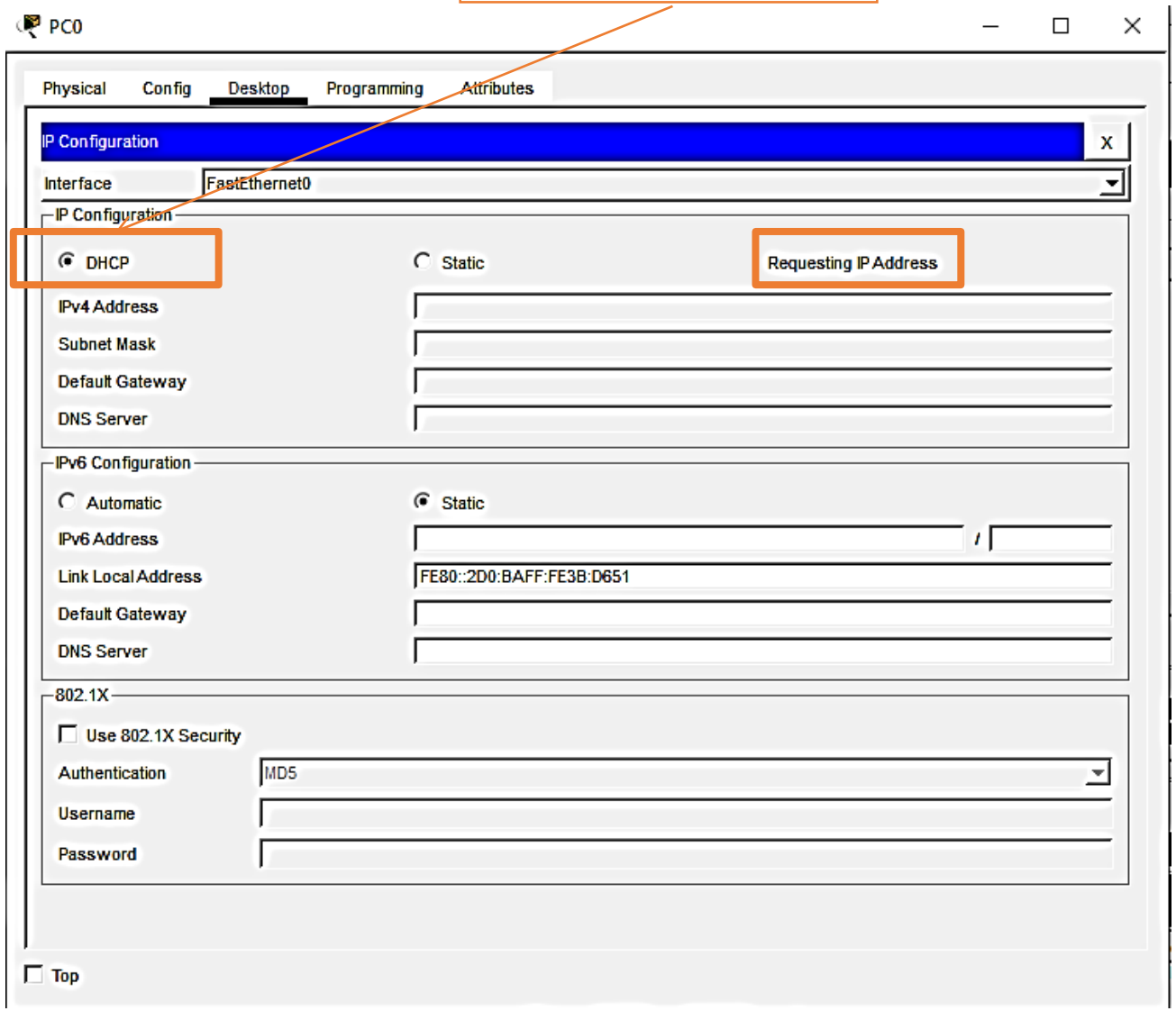
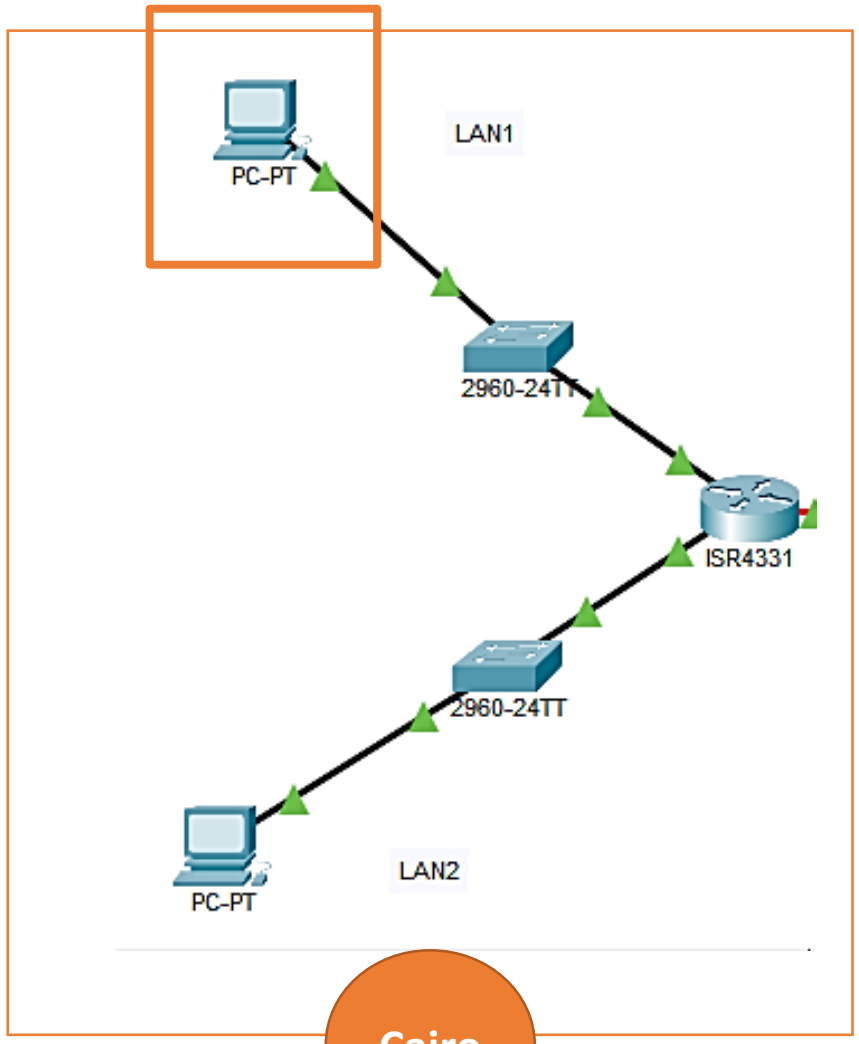
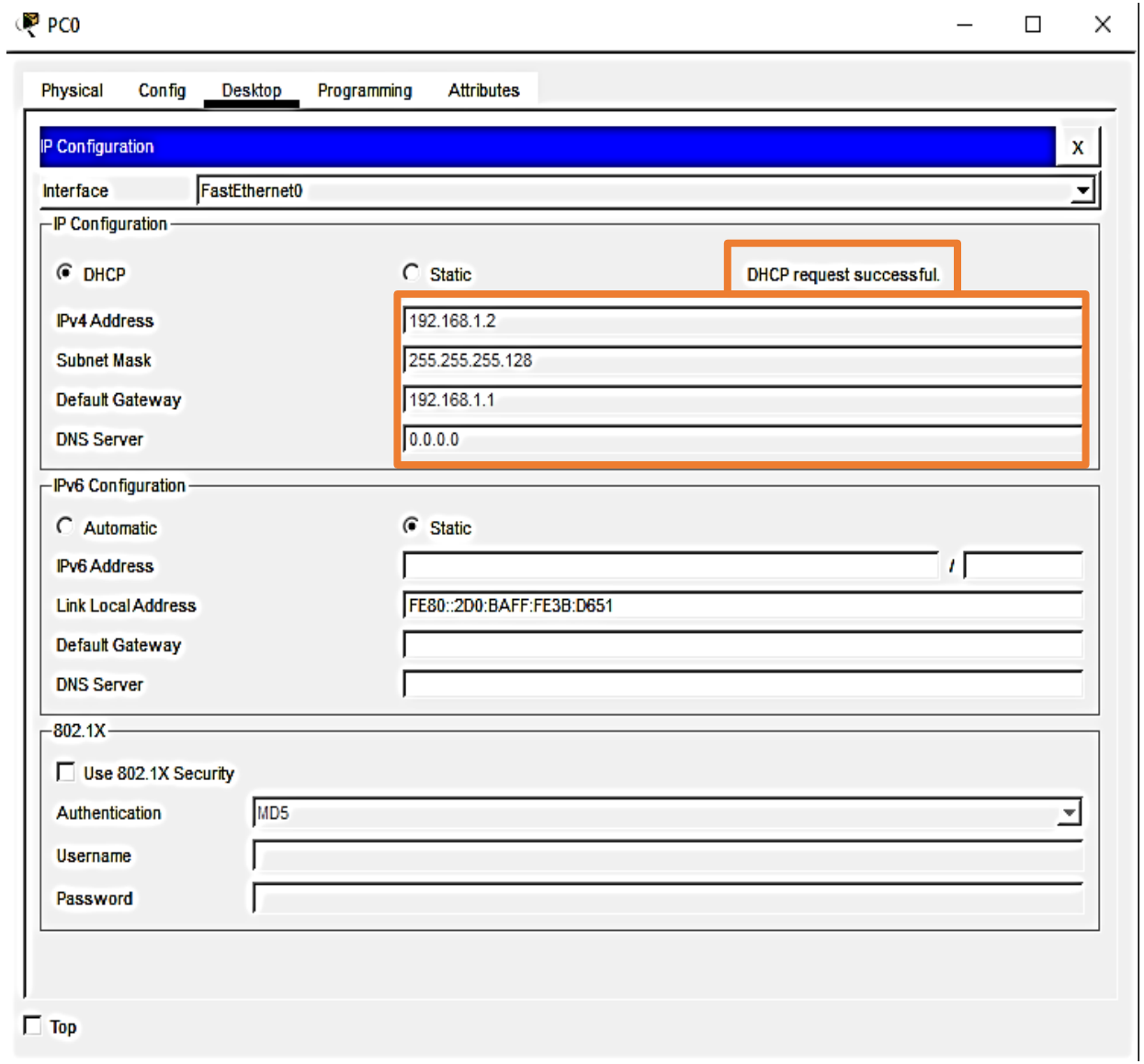


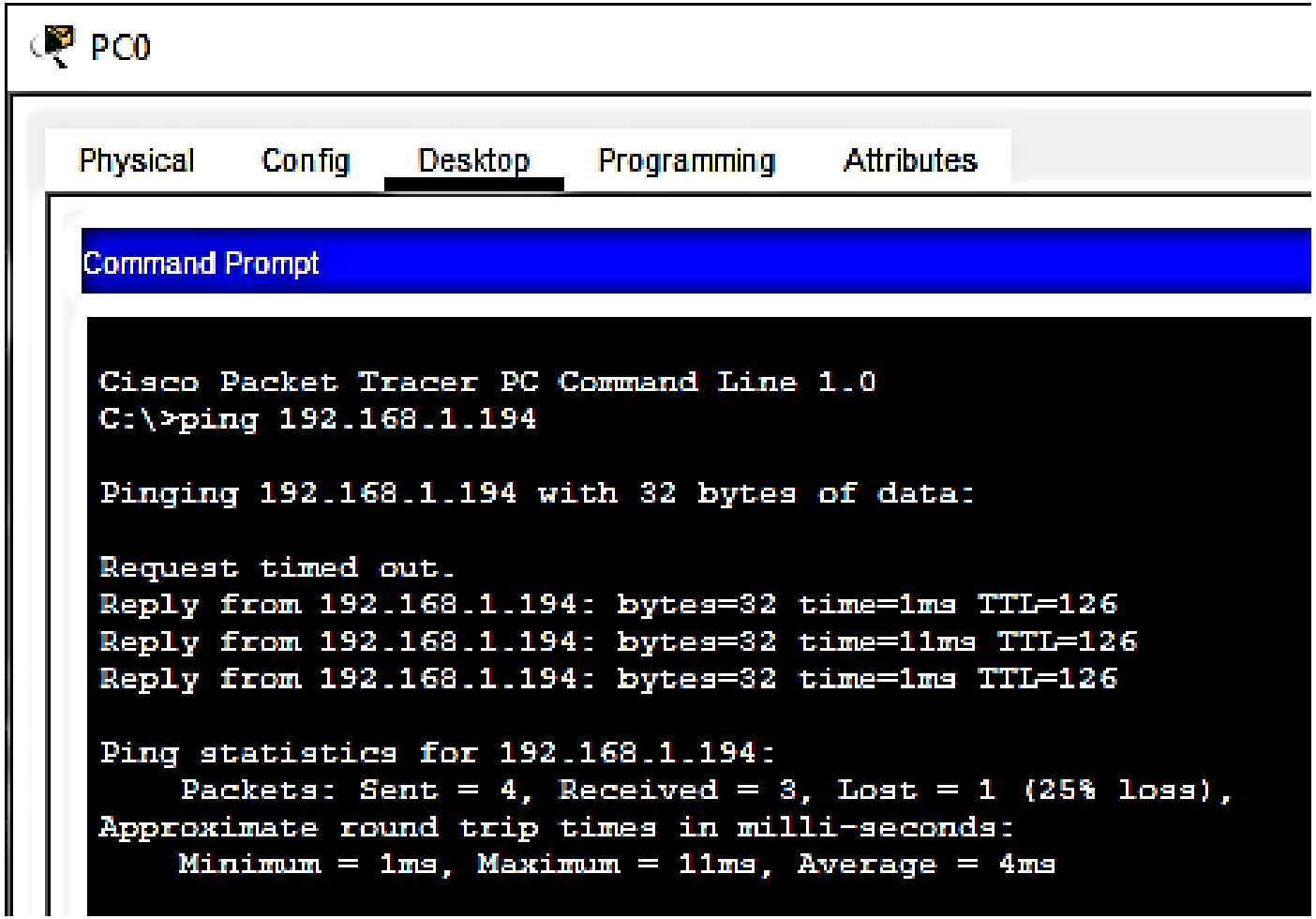
Configure PCs





Test Network

- Open command prompt on a PC and test:



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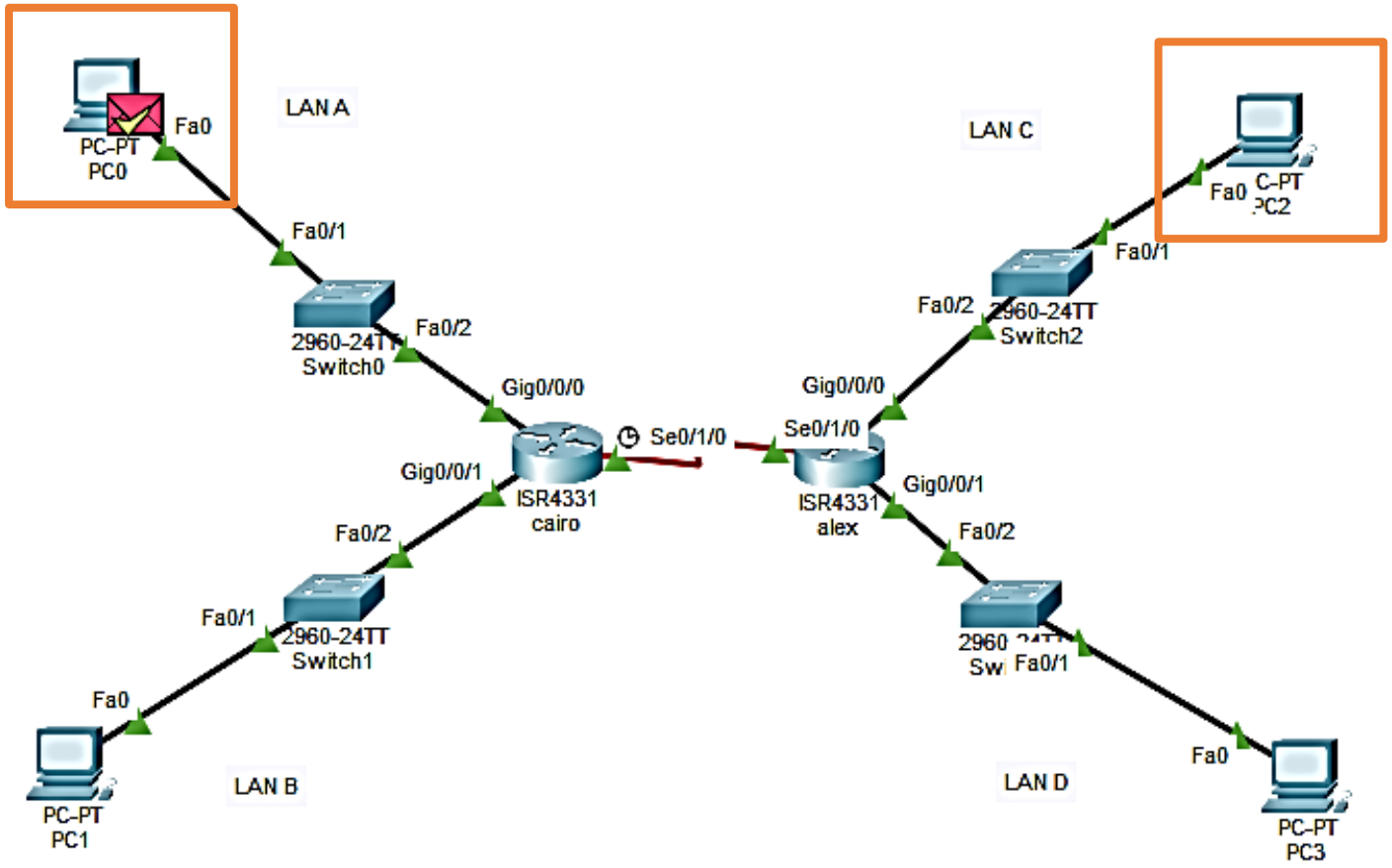
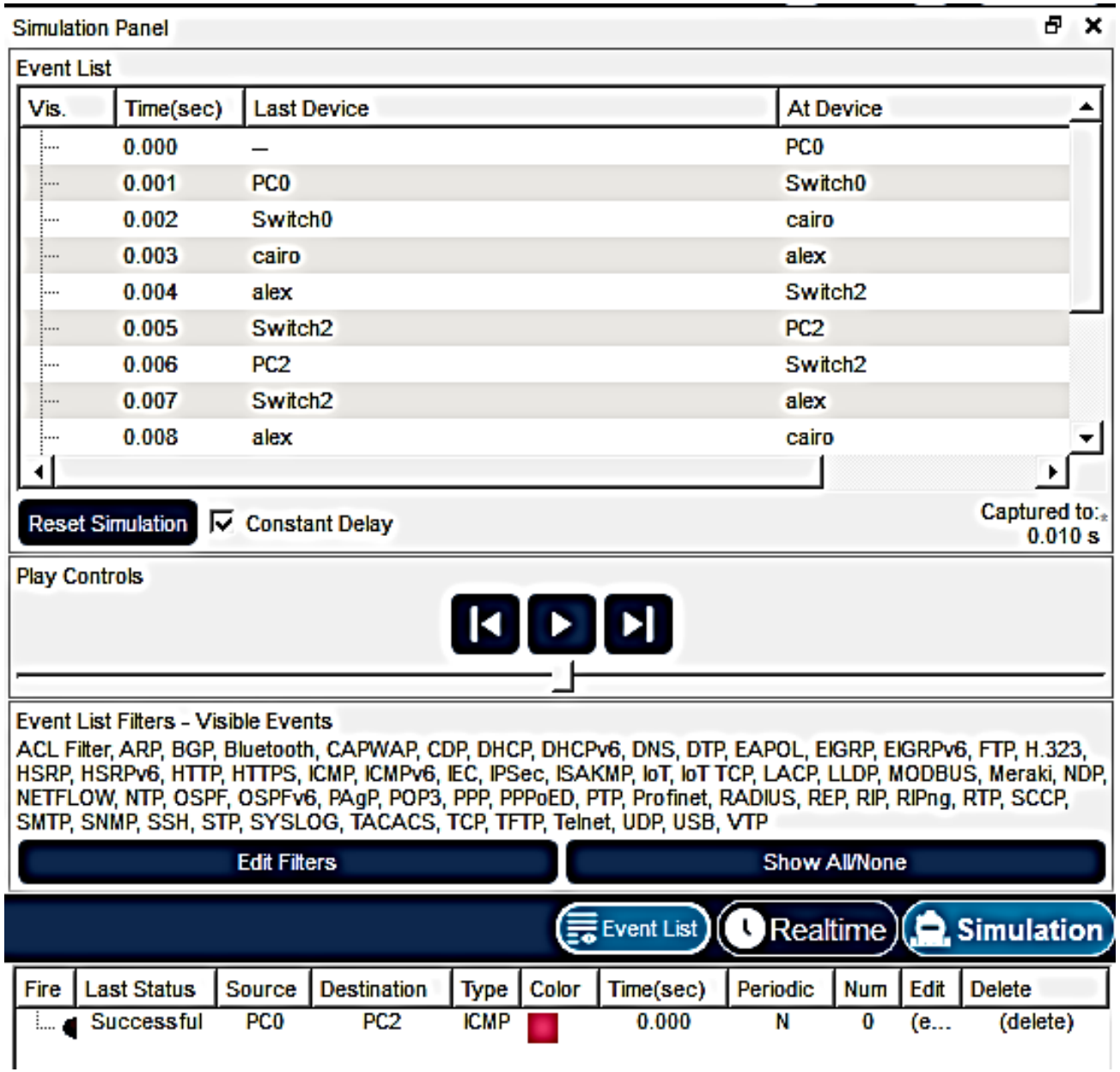
- You can send a message from any PC to another by adding simple PDU as figure:



Simulation Info

The Simulation Panel helps you:

- See how packets travel between devices
- Understand network protocols
- Troubleshoot network problems
- Observe processes like DHCP, ARP, and Ping.



Any Questions?

Thanks ☺